Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/22/2018
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Project Title:	Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Poulouse, Shibu Ph.D. (Tufts University School of Medicine) Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville, MD)		
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Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to single, above threshold doses of HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neuronal and behavioral functioning after the conclusion of the mission. The effects of exposure to multiple, subthreshold doses of the same or different particles or protons have not been extensively studied. The objectives of the proposed research program are: (1) to evaluate the effects of repeated below threshold/above threshold exposures to the same and different HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our data indicate that there is a significant risk that exposure to low, NASA-relevant doses of HZE particles produces alterations in central nervous system (CNS) function and in the behaviors that are dependent upon the integrity of specific CNS systems. However, current research suggests several caveats that might modulate the effectiveness of such exposures in disrupting neuronal function and cognitive performance. The experiments presented in this report were designed to evaluate several factors which may affect the degree to which exposure to space radiation might affect the cognitive ability of astronauts on exploratory class missions. The first two series of experiments utilize exposure to 4He particles because 4He and 1H particles will constitute the major source of radiation within the spacecraft during exploratory class missions. One series of experiments examines the effects of repeated exposure to a below threshold dose of 4He particles on a subsequent exposure to a second dose which is either below or above threshold. This experiment attempts to mimic the most likely pattern of radiation exposure to which astronauts will be exposed on exploratory class missions and involves repeated exposures to below threshold doses. In this context, threshold is defined as the lowest dose of a specific particle that will produce a deficit in cognitive performance. These experiments presented in the current report using both subthreshold and above threshold doses of 4He particles suggest that the effects on cognitive performance may be task specific and may also vary as a function of the interval between exposures. Current experiments (6 month interval) combined with previous research (2, 24, or 72 hr intervals) suggest that a below threshold conditioning exposure can either enhance or prevent the effects of exposure to 4He particles depending upon the time interval and specific cognitive task. These experiments are currently being repeated in order to establish the validity and reliability of the results. A second series of experiments evaluated the effects of exposure to HZE particles on the performance of female organisms and the role of estrogen as a mediator. Ovariectomized (OVX) female rats implanted with tubes containing estradiol or vehicle and intact rats have been exposed to 4He particles. The results suggested that the presence of estrogen at the time of irradiation functions to exacerbate the disruptive effects of exposure to 4He particles on cognitive performance. In contrast, intact female rats and OVX rats given implants of the vehicle without estrogen showed improved responsiveness to changes in reinforcement contingencies. The mechanisms that may be responsible for the disrupted performance of the estrogen-treated subjects are not certain. We plan to evaluate the effectiveness of anti-oxidant diets as mediators of these effects. The third series of experiments with rats was designed to determine whether the disruption of neuronal function and cognitive performance following exposure to HZE particles resulted from direct effects of the particles on neurons (“hits”) or whether the neuronal and cognitive effects were due to indirect effects of oxidative stress and neuroinflammation, which can be produced by the effects of HZE particles throughout the entire body. If oxidative stress and neuroinflammation are the critical mediators there should be no differences in the behavioral or neuronal endpoints as a function of the location of the exposure (head-only or body-only). If direct effects on neurons mediate the differences in the effects of higher or lower linear energy transfer (LET) particles, we should see threshold differences as a function of particle LET with head-only, but not with body-only exposures. While neurochemical analyses are ongoing, preliminary data on cognitive performance indicates that there are no differences in the effects of head-only or body-only exposures, a result that would be consistent with the hypothesis that the primary mechanism leading to a disruption of cognitive performance following exposure to HZE particles involves radiation-induced oxidative stress and neuroinflammation.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Carrihill-Knoll KL, Miller MG, Shukitt-Hale B. "Age as a factor in the responsiveness of the organism to the disruption of cognitive performance by exposure to HZE particles differing in linear energy transfer." Life Sci Space Res (Amst). 2018 Feb;16:84-92. Epub 2017 Dec 13. https://doi.org/10.1016/j.lssr.2017.12.001 ; PubMed PMID: 29475524 , Feb-2018